

Multi-class MI-EEG Classification: Using FBCSP and Ensemble Learning Based on Majority Voting

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Abstract—How to decode multi-class motor imagery EEG and decipher an individual's intents more accurately has always been a problem worthy of study in biomedical field. In the process of recognition and classification of 2-class motor imagery EEG, the Common Spatial Pattern (CSP) algorithm detects ERD /ERS by calculating the spatial filter and achieves great success. In this paper, we use self-frequency Segmented Filter Bank Common Spatial Patterns (FBCSP) with the one versus one (OVO) multi class extension on the classification of 4-class MI-EEG (BCI Competition IV Datasets 2a), and the Majority Voting strategy is adopted for the individual classifiers we select. We finally get a considerable classification result and the comparison of different methods shows that the performance of ensemble learning based on voting strategy is better than that of using individual classifier.

Keywords—multi-class motor imagery EEG, OVO-FBCSP, ensemble learning, majority voting

I. INTRODUCTION

Human scalp Electroencephalogram (EEG) can be divided into endogenous and exogenous. Exogenous EEG, such as Event Related Potential (ERP) and Steady State Visual Evoked Potential (SSVEP), usually needs specific external stimulation to induce. However, unlike exogenous EEG signals, endogenous spontaneous EEG signals come from the subjects' own independent imagination or action, such as the related rhythmic changes in the motor sensory area during the motor imagery task — Event Related Synchronization and Event-Related Desynchronization (ERS/ERD). Compared with the two kinds of signals, ERS / ERD can provide spontaneous control modes without external stimulation. The BCI system based on motor imagery can directly transform human imagination or intention into instructions or commands for the control of peripheral devices, and it does not need to rely on the peripheral nerves and muscles of the brain. It provides not only a new but also convenient paradigm for human-computer interaction [1][2].

When we imagine a specific limb movement, in a specific brain area, the EEG signal will produce some specific rhythm increase or decrease in a specific frequency band. Therefore, for the feature extraction of motor imagery EEG signal, it is particularly effective to capture the spatial features. Among them, the Common Spatial Pattern algorithm is accepted as an algorithm that can obtain higher recognition accuracy, that

is, the corresponding spatial features can be obtained by calculating the spatial filter. CSP was first proposed by Z.J. Koles in 1990 and applied to feature extraction of EEG [3]. On the basis of CSP algorithm, many algorithms are proposed to improve the performance of MI-EEG feature extraction and recognition, such as the Sub-band Common Spatial Pattern (SBCSP) algorithm proposed by Novi Q [4], Filter Bank Common Spatial Pattern (FBCSP) algorithm proposed by Kai K Ang [5], etc.

The features extracted from MI-EEG will be sent to the trained classifier. Among the traditional classifiers, there are many classifiers that are most frequently used, such as Linear Discriminant Analysis (LDA) algorithm, Logical Regression (LR) algorithm, Support Vector Machine (SVM) algorithm, etc. the subsequent ensemble classifiers based on traditional classifiers can also achieve considerable classification results [6].

In this paper, a voting method based on OVO-FBCSP and multiple classifiers is proposed for feature extraction and classification of multi-class motor imagery EEG signals, the FBCSP is different from [5], we choose to combine sub-features directly, and the frequency band is self-segmented. 2-class imagery datasets are used to verify the voting method Firstly. Different ensemble classifiers and traditional individual classifiers are used to classify the 2-class motor imagery features, the results show that the ensemble learning based on voting algorithm performs better than other classifiers.

Therefore, in the research of 4-class MI-EEG classification, we combine the Majority Voting algorithm with one versus one (OVO) multi-class extended FBCSP, and use five common algorithms as the individual classifiers for voting. In the voting process, the results predicted by the worst classifier are removed, and the remaining four prediction results are voted. If the remaining four results are different in the voting process, the results predicted by the classifier with the most outstanding average classification effect are selected.

II. METHODS

A. Common Spatial Pattern

Common space pattern algorithm (CSP) is a feature extraction algorithm for classification for 2-class problem, and it is recognized as one of the most widely used algorithms that

can obtain a high recognition accuracy [7-11]. The main idea 8joint diagonalization of the covariance matrix to calculate the required optimal spatial filter. Then the original signal projection is able to make a kind of signal variance reach maximum while another kind of signal variance reach minimum, so that the two types of signal variances get the largest difference. Furthermore, the feature vectors are constructed by the variances of the signals and sent to the classifier for classification and recognition.

Taking left-right motor imagery tasks as an example, it is assumed that the subscript L/R represents the left hand and the right hand respectively, and the corresponding EEG signals are E_L and E_R .

The covariance matrices after normalization of E_L and E_R are:

$$C_L = \frac{E_L E_L^T}{\text{trace}(E_L E_L^T)} \quad (1)$$

$$C_R = \frac{E_R E_R^T}{\text{trace}(E_R E_R^T)} \quad (2)$$

where the E^T represents the transpose form of signal E, and $\text{trace}(E)$ represents the sum of diagonal elements of matrix E.

$\overline{C_L}$ and $\overline{C_R}$ represents the mean covariance of the two classes of left and right hand, then the mixed spatial covariance can be obtained as:

$$C_M = \overline{C_L} + \overline{C_R} \quad (3)$$

Eigenvalue decomposition of the mixed space covariance matrix C_M is performed to obtain (4):

$$C_M = U_M \Lambda_M U_M^T \quad (4)$$

In (4), the diagonal matrix Λ_M constituted by the eigenvalues of the matrix C_M , U_M represents the eigenvector matrix corresponding to the eigenvalues.

Then construct the whitening matrix A according to the diagonalized result:

$$A = \sqrt{\Lambda_M} U_M^T \quad (5)$$

The matrices of $\overline{C_L}$ and $\overline{C_R}$ transformed by whitening matrix are expressed as:

$$T_L = A \overline{C_L}^T A^T, T_R = A \overline{C_R}^T A^T \quad (6)$$

Then the eigenvalue decompositions of the matrices are obtained as:

$$T_L = B_L \Lambda_L B_L^T, T_R = B_R \Lambda_R B_R^T \quad (7)$$

From (7), it is not difficult to get the relation between T_L and T_R matrices that the sum of eigenmatrices is the identity matrix, and at the same time, eigenvectors are equal:

$$\Lambda_L + \Lambda_R = I \quad (8)$$

$$B_L = B_R = B \quad (9)$$

The eigenvalue matrices corresponding to T_L and T_R in (7) are arranged in descending order and ascending order respectively, which is the best for separating the variances of the two signal matrices. The first n rows and the last n rows of the eigenvector matrix are selected to form a new matrix, and the optimal spatial filter is obtained as:

$$W_{2ns} = B_{2n}^T A \quad (10)$$

The spatial filter is used to project the original signal, and the projected signal is expressed as:

$$P_{2n} = W_{2ns} A \quad (11)$$

The eigenmatrix of the signal after projection is calculated. The eigenmatrix is f , and the row vector of the matrix is f_i , which can be obtained from:

$$f_i = \log \left(\frac{\text{var}(P_i)}{\sum_{k=1}^{2n} \text{var}(P_k)} \right) \quad (12)$$

In (12), $\text{var}(P)$ is the variance of the matrix P , and the required eigenmatrix f is obtained by combining the row vectors f_i of $i=1,2,...,2m$.

B. Filter Bank Common Spatial Pattern

On the basis of CSP, FBCSP uses multiple band-pass filters and spatial filters. It decomposes the original signal into signals on multiple frequency bands after passing through N band-pass filters with the same bandwidth.

For the FBCSP algorithm adopted in this paper, we modified the manual frequency setting method like [12], the maximum and minimum values of the overall frequency range as well as the number of sub-bands are set to carry out isometric self-segmentation of the frequency band. For seven sub-band signal we set at first, the corresponding seven spatial filters are calculated by CSP algorithm, then the original signal is divided into sub-bands for spatial filtering by the method described above and multiple features are calculated corresponding to each sub-band. The spatial features of sub-bands extracted were selected or combined, and the final feature matrix was sent to classifiers for training.

C. Ensemble Learning Based on Majority Voting

Ensemble learning is a learning task that combines the decisions of many single learners and aggregates the predicted results. In most cases, ensemble learning based on multiple predictors is better than a single predictor.

Bagging (Bootstrap-aggregation) reduces overfitting by averaging the prediction of the randomly generated subsets of data sets. For example, Random Forest (RF) is an algorithm based on Bagging [13]. Boosting is an ensemble learning method that reduces the bias through cascading training of classifiers. And it tries to increase the weight of incorrectly classified samples in every iteration, so as to improve the probability of samples with difficulty in classification to be selected in the subsequent classifiers [14]. Voting needs to train multiple classifiers and votes on the predicted results of multiple classifiers to choose the best predicted results.

In the classification process of the 2-class of MI-EEG, we establish multiple individual learner models, and compare the Bagging based on Random Forest algorithm, the Boosting, and the Majority Voting algorithm. It is concluded that voting is superior to other classifiers, so in the classification process of the 4-class of MI-EEG, on the basis of choosing a better feature extraction algorithm, we used Majority Voting algorithm to aggregate the results of individual classifiers. The overall classification pipeline is shown in Fig.1.

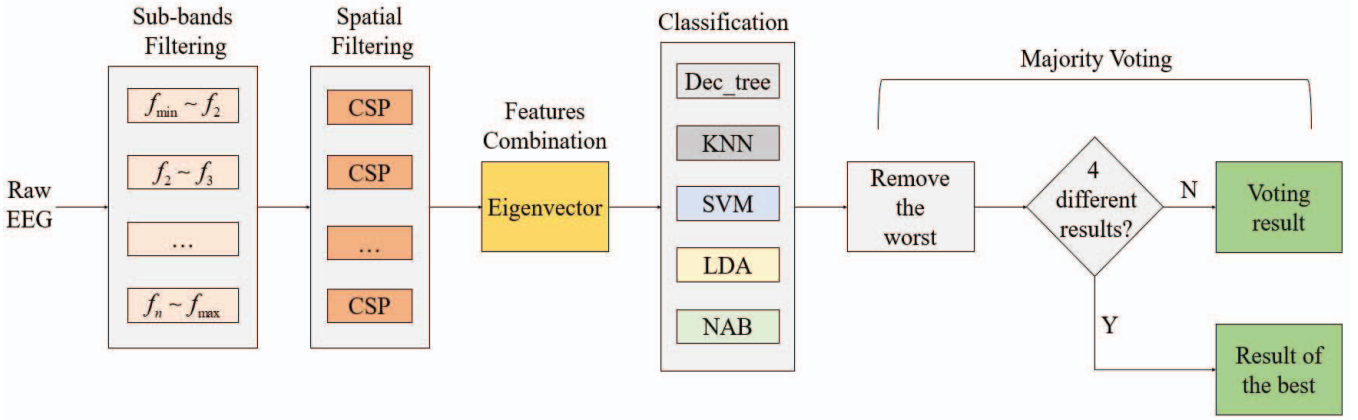


Fig.1 Classification pipeline based on FBCSP and Voting

Firstly, the overall minimum frequency and maximum frequency as well as the number of sub-bands are set in the code, and a Chebyshev II filter bank is used to conduct bandpass filtering on the original EEG data. Different from [5], the spatial features obtained by spatial filtering of sub-bands data are combined in this paper to obtain the final feature matrix. Five individual classifiers including Decision Tree, KNN, SVM, LDA and Naive Bayes were trained by using the feature matrix of training set. The feature matrix of test data was put into the trained models for classification. The classifier model with the best performance was selected, and the one with the worst effect was removed.

After the selection and removal of the models, there will be four results for each prediction, and the results will be voted. If four different results are obtained and it is impossible to judge, the prediction result of the optimal model will be selected.

Finally, the classification performance obtained by different classification methods are computed and compared.

III. DATASETS AND RESULTS

In the verification of the MI-EEG classification in this paper, we used two common datasets: the BCI Competition III IVb dataset for 2-class classification and the BCI Competition IV 2a for 4-class classification.

A. Dataset A: 2-Class dataset

Before the classification of the 4-class MI-EEG, the classification experiment of the 2-class MI-EEG was first conducted using the BCI Competition III IVb dataset to verify the ensemble learning based on Voting.

Seven sessions make this dataset up, the subject of the experiment sat in a suitable chair and put arms on the armrests. Also, the premise is that the subjects are healthy. During the first three sessions of the task, visual cues were presented to the subjects in the form of 3.5s of letters, including L for left hand, F for right feet, and Z for tongue, the time of relaxation varies from 1.75 to 2.25s. The latter four sessions include tasks without visual cues, the target classes left, foot and relax were ordered by acoustic stimuli for periods for varying length between 1.5 and 8 seconds. Intermitting periods were given as above.

The number of sampling channels in the dataset is 118, and the EEG signals are band-pass filtered between 0.05 and

200Hz. The experimental paradigm corresponding to the data set is shown in Fig.2.

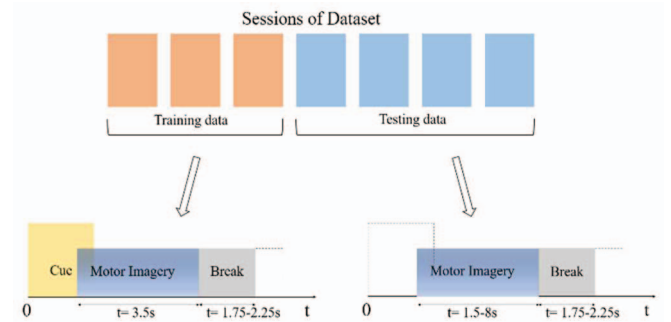


Fig.2 Experiment paradigm of BCI Competition III IVb dataset

B. Dataset B: 4-Class dataset

After two MI-EEG classification methods were validated, we performed 4-class classification experiments using the BCI Competition IV 2a dataset with an optimized feature extraction algorithm.

The dataset consists of the MI-EEG of 9 subjects. The cue-based BCI paradigm contains four types of MI tasks, namely, left hand, right hand, feet and tongue. EEG data of each subject were recorded for one session within two days respectively, which were used as training data and test data respectively. Each session consists of 6 runs with a short break between each run. Each run consists of 48 imagination tasks (12 trails for each class). Each session contains 288 experimental trails.

The timing of a trial is as follows. The subject could catch a cross and receive a brief cue tone on the screen while $t = 0s$. Then one of the different imagination tasks cue (actions of left hand, right hand, foot and tongue) would appear on the screen while $t = 2s$, which would be presented for 1.25s as arrows in the left, right, down and up directions. The subject should start the thinking task while caught the cue on the screen, and stop thinking while there was no cross at $t = 6s$. The last stage was taking a quick rest and no feedback. Then subject should prepare for the next trial.

The number of sampling channels in the dataset is 22, the sampling frequency is 250Hz, and the EEG signals are bandpass filtered between 0.5-100Hz. The montage of electrode position distribution and the timing scheme of the

paradigm corresponding to the data set are shown in Fig. 3 and Fig. 4. For more details, please refer to the document [15].

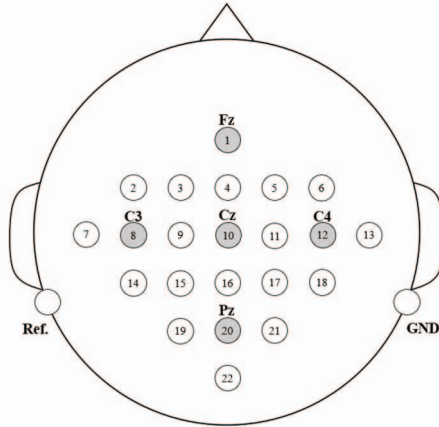


Fig.3 Electrode montage of BCI Competition IV 2b dataset

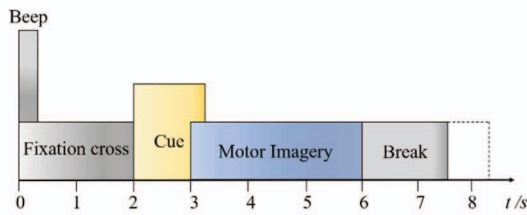


Fig.4 Experiment paradigm of BCI Competition IV 2b dataset

C. Results

In this section, we present the classification results of motor imagery EEG signals on the datasets using our proposed method, including the classification of 2-class MI-EEG and the classification of 4-class MI-EEG.

In the classification of the 2-class MI-EEG, the data of the first three sessions were used for training, and the data of the last four sessions were used for testing. Different individual classifiers and ensemble classification methods were used to compare the results. Fig.5 shows the classification accuracies of the test set data under different classification methods. Fig.6 is the radar map of Kappa values in the corresponding method.

By comparing different traditional classifiers and ensemble classifier methods, we conclude that the SVM of RBF kernel function has the best performance in individual classifier, even better than ensemble classifier. Under the condition of selecting the appropriate number and type of individual classifiers, the ensemble algorithm based on voting has better classification effect than all other classifiers.

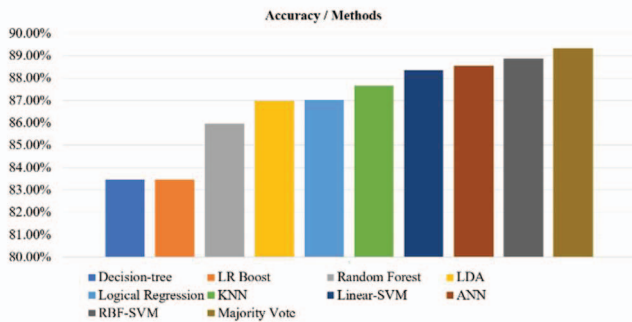


Fig.5 Comparison of 2-class MI-EEG classification

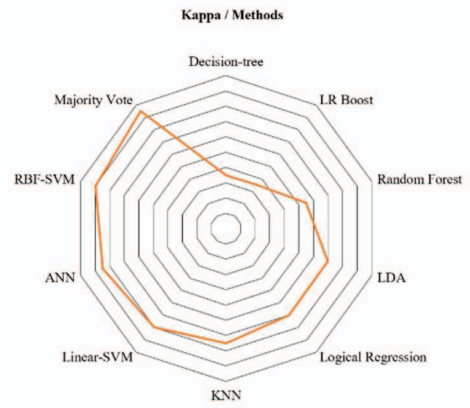


Fig6. Radar map of Kappa/Methods in 2-class MI-EEG classification

In the classification of 4-class motor imagination EEG signals, the EEG data of 9 subjects in the dataset were labeled T for training while the data were labeled E for testing.

In the classification of 4-class, we adopt the superior feature extraction algorithm FBCSP with the one versus one (OVO) and the one versus rest (OVR) multi class extension respectively to process the data. Here, the classification effect obtained under the OVO-FBCSP classification strategy is better, so we verified the Voting algorithm under the OVO-FBCSP method, and the results show that the Voting method is indeed superior to the single classifiers, with the highest classification accuracy reaching 82.99%. Table1 and Table2 respectively show the classification effects of different classifiers in two situations.

TABLE I. PERFORMANCE OF 4-CLASS MI-EEG UNDER OVR-FBCSP

Sub.	OVR-FBCSP				
	<i>Dec_tree</i>	<i>KNN</i>	<i>LDA</i>	<i>SVM</i>	<i>Naive Bayes</i>
1	64.85	60.07	73.61	72.22	72.92
2	46.88	51.04	58.33	60.76	60.76
3	65.97	64.24	75.69	76.39	75.35
4	60.07	53.13	66.67	66.32	64.58
5	32.64	35.07	48.95	51.39	50.00
6	38.54	37.85	44.79	46.53	43.75
7	50.69	64.24	68.75	70.14	74.31
8	55.21	54.17	67.71	72.22	76.04
9	49.31	44.44	61.46	65.28	64.93
Avg.	51.54	51.58	62.89	64.58	64.74

TABLE II. PERFORMANCE OF 4-CLASS MI-EEG UNDER OVO-FBCSP

Sub.	OVO-FBCSP					
	<i>Dec_tree</i>	<i>KNN</i>	<i>Naive Bayes</i>	<i>LDA</i>	<i>SVM</i>	<i>Voting</i>
1	68.75	76.39	81.25	81.94	81.94	81.94
2	46.18	55.21	57.99	56.94	57.99	57.99
3	78.47	75.69	73.69	73.96	75.69	76.74
4	50.69	50.74	54.51	57.99	64.24	64.24
5	52.78	55.21	59.03	61.11	62.85	61.81
6	35.42	39.24	41.67	44.44	45.14	44.44
7	71.18	72.92	75.00	79.17	81.94	82.99
8	71.88	70.49	76.74	71.18	77.08	76.74
9	59.38	59.38	68.40	68.85	69.44	69.79
Avg.	59.14	61.81	65.40	66.17	68.48	68.52

IV. CONCLUSION

In this paper, we do feature extraction and recognition on both 2-class and 4-class MI-EEG. In the 2-class MI-EEG, we use various traditional single classifiers and ensemble classification methods to classify the dataset. It is concluded that under the condition of selecting the appropriate number and types of classifiers, the Majority Voting method can improve the classification accuracy of the data, and its average accuracy is higher than that of all single learners. On this basis, 4-class MI-EEG are further classified. OVO-FBCSP is used to extract features from the data, and five traditional classifiers are used for classification. After judging the best and worst, the proposed voting strategy is used for voting classification, and finally get the result that the performance of this method is better than that of all individual classifiers.

This classification experiment has achieved considerable results in the classification of MI-EEG. In the future work, we will process the extracted features more appropriately in FBCSP algorithm, and consider selecting better individual classifiers for classification and recognition.

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